DO NOW, AO1 REVISION

Moments, levers and gears

Forces
Paper 2, Separate Physics only
11 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 3 starred (★) questions first.

	QUESTION	ANSWER	✓
1	**State what is required to cause an object to rotate.**	A force that does not act through the pivot or centre of mass.	☐
2	**Name the turning effect of a force.**	A moment	☐
3 ★	**State which distance is used when calculating the moment of a force.**	The perpendicular distance from the pivot to the line of action of the force	☐
4	**The clockwise and anticlockwise moments about a pivot are equal. State what this tells you about the object.**	It is balanced.	☐
5 ★	**Explain how a lever can reduce the force needed to turn something.**	Pushing further from the pivot gives the same moment with a smaller force.	☐
6	**State the function of a lever.**	To transmit the rotational effect of a force.	☐
7	**State the function of a simple gear system.**	To transmit the rotational effect of a force.	☐
8	**State the equation linking moment, force and distance.**	moment = force × perpendicular distance from the pivot (M = F d)	☐
9	**State the standard unit of the moment of a force.**	Newton-metre (N m)	☐
10 ★	**State the principle of moments for a balanced object.**	Total clockwise moment about the pivot = total anticlockwise moment.	☐
11	**A small gear drives a larger gear. State the effect on the speed of rotation and on the moment.**	The larger gear turns more slowly but with a bigger moment.	☐

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